

7. Developing Responsive User Interface

7.1 Add containers to the Oracle JET web application

A **responsive layout** adjusts automatically to fit **any screen size** — from mobile phones to large monitors.

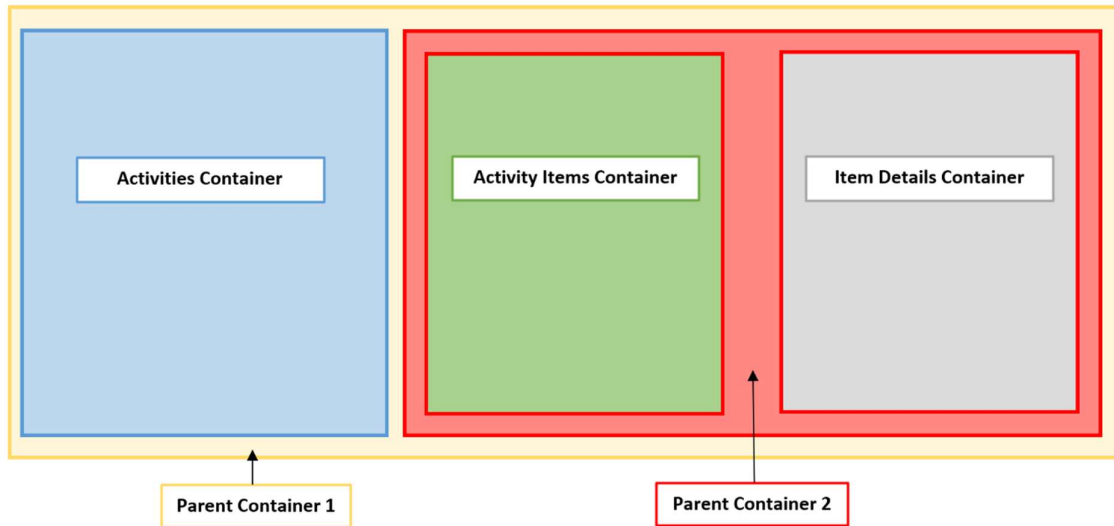
In Oracle JET:

- Use `**oj-flex**` and `**oj-flex-item**` classes to arrange elements in **rows or columns**.
- These elements **resize** as the screen size changes — they **grow or shrink** to fit the space.
- Add borders and background color for layout clarity using `**oj-panel**` and `**oj-panel-bg**`. (These are **just for visuals** and can be removed later.)

App Structure (What you'll build):

- A **main container** → holds everything.
 - Inside it:
 - A **second container** → holds:
 - **Activity Items**
 - **Item Details**
 - And an **Activities container** separately.

This structure helps organize the UI clearly and keeps it responsive across devices.



Objectives

In this tutorial, you will add containers and panels to convert a page layout to a responsive page layout in an Oracle JET web app by using Oracle JET layout and style classes.

Prerequisites

- A development environment set up to create Oracle JET web apps that includes an installation of Node.js
- Google Chrome web browser installed and set as the default web browser
- Access to the [Oracle JET Developer Cookbook](#)
- (Option 1) Completion of the final tutorial in the previous learning path in this series: **Prepare to Deploy an Oracle JET Web Application** (from chapter 5)

Task 1 : Add Activities Containers

You are going to **modify your Oracle JET web app layout** using **Flexbox** to make it **responsive** and organized.

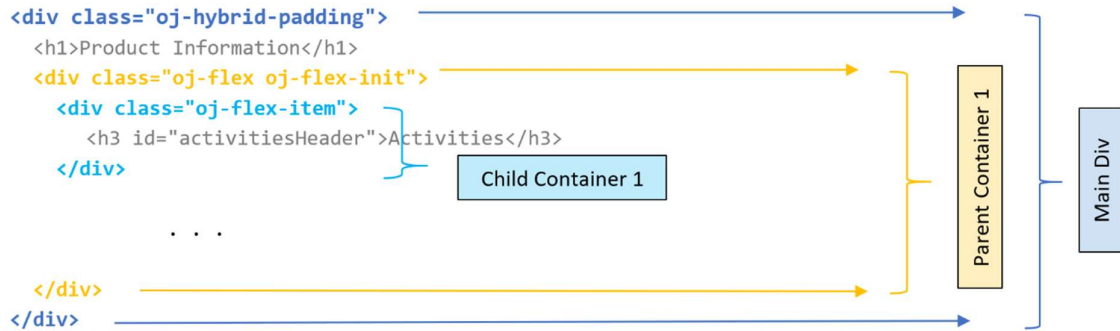
What to Do:

- Use `<div class="oj-flex">`

This creates the **Parent Container 1** (flex container) that holds all content inside.

- Add `<div class="oj-flex-item">` inside

This creates **Child Container 1**, where you'll place the **Activities list**.



1. Navigate to the JET_Web_Application/src/tj/views directory and open the dashboard.html file in an editor.
2. Find the Product Information heading h1 element, and beneath it add an outer div element with the id attribute value parentContainer1 along with the Oracle JET container class. This div element encloses all other containers within it. Add a closing </div> tag above the last closing </div> at the end of the file.

```

<div class="oj-hybrid-padding">
  <h1>Product Information</h1>
  <!-- Parent Container 1 contains all the panels in the app -->
  <div id="parentContainer1" class="oj-flex oj-flex-init">
    <div id="activitiesContainer">
      <h3 id="activitiesHeader">Activities</h3>
      . . .
    </div>
  </div>
</div>

```

3. Find the div element where id="activitiesContainer", before the Activities heading, and add an Oracle JET container item class

```

<!-- The Activities container displays the Activities list -->
<div id="activitiesContainer" class="oj-flex-item">
  <h3 id="activitiesHeader">Activities</h3>
</div>

```

4. Save the dashboard.html file. Your code should look similar to below html code

```

<div class="oj-hybrid-padding">
  <h1>Product Information</h1>
  <!-- Parent Container 1 contains all the panels in the app -->
  <div id="parentContainer1" class="oj-flex oj-flex-init">
    <!-- The Activities container displays the Activities list -->

```

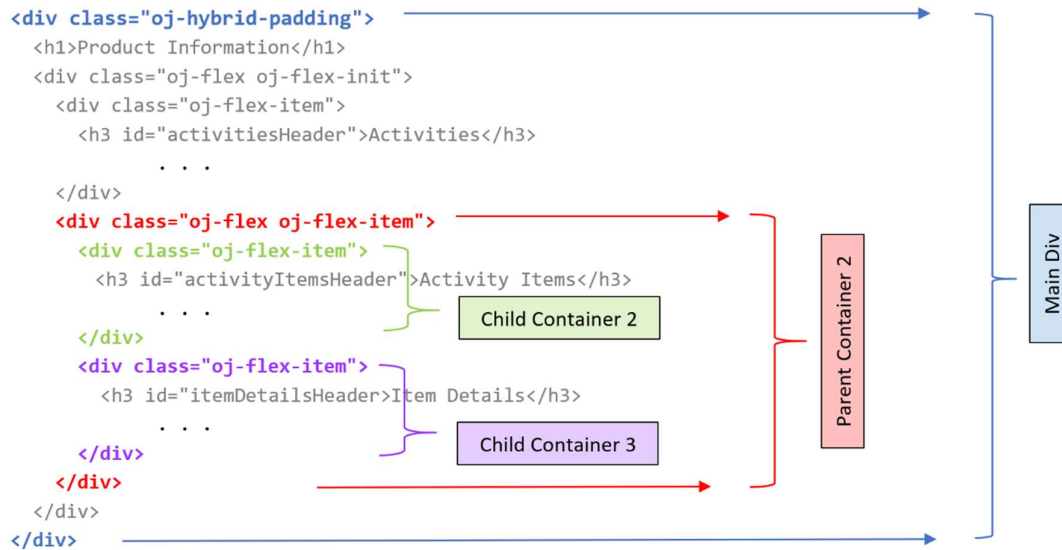
```

<div id="activitiesContainer" class="oj-flex-item">
  <h3 id="activitiesHeader">Activities</h3>
  <oj-list-view id="activitiesList" aria-
labelledby="activitiesHeader" data="[[activityDataProvider]]"
    gridlines.item="visible">
    <template slot="itemTemplate">
      <div>
        <strong>
          <oj-bind-text value="[[ $current.data.name]]"></oj-bind-
text>
        </strong>
      </div>
    </template>
  </oj-list-view>
</div>
<div id="itemDetailsContainer">
  <h3>Item Details</h3>
  <oj-label for="basicSelect">Select Chart:</oj-label>
  <oj-select-single id="basicSelect" class="selectSingleStyle"
data="[[chartTypesDP]]" value="{{val}}">
    </oj-select-single>
    <oj-chart id="barChart" type="[[val]]"
data="[[chartDataProvider]]" animation-on-display="auto"
    animation-on-data-change="auto" hover-behavior="dim"
class="chartStyle">
      <template slot="itemTemplate" data-obj-as="item">
        <oj-chart-item value="[[item.data.value]]" group-id="[[
[item.data.group] ]]"
          series-id="[[item.data.series]]">
        </oj-chart-item>
      </template>
    </oj-chart>
  </div>
</div>
</div>

```

Task 3: Add Item Details Containers

1. Modify the Oracle JET web app responsive layout to add the Item Details containers. Parent Container 2 contains the two child containers. Child Container 2 is left empty but in the later tutorials will contain the Activity Items list in it. Child Container 3 contains the Item Details chart.



2. Find the div element where id="itemDetailsContainer". Above it, add an outer div element with an id attribute with the value parentContainer2, as well as an Oracle JET container class and container item class. Add a closing </div> tag above the second-to-last closing </div> in the file.

```

<!-- Parent Container 2 surrounds the Item Details container -->
<div id="parentContainer2" class="oj-flex oj-flex-item">
  <div id="itemDetailsContainer">
    <h3>Item Details</h3>
    . . .
  </div>
</div>
</div>
</div>

```

3. Underneath the div element where id="parentContainer2", add an empty div element with an id attribute with the value activityItemsContainer, as well as an Oracle JET container item class.

```

<!-- Parent Container 2 surrounds the Activity Items and
Item Details child containers -->
<div id="parentContainer2" class="oj-flex oj-flex-item">
  <div id="activityItemsContainer" class="oj-flex-
item"></div>
  <div id="itemDetailsContainer">
    <h3>Item Details</h3>

```

```

        . . .
    </div>
</div>

```

4. Finally, add an Oracle JET container item class to the div element where id="itemDetailsContainer".

```

<!-- Item Details container displays a chart upon selection
-->
<div id="itemDetailsContainer" class="oj-flex-item">
    <h3>Item Details</h3>
</div>

```

5. Save the dashboard.html file. Your code should look similar to the below code

```

<div class="oj-hybrid-padding">
    <h1>Product Information</h1>
    <!-- Parent Container 1 contains all the panels in the app -->
    <div id="parentContainer1" class="oj-flex oj-flex-init">
        <!-- The Activities container displays the activities list -->
        <div id="activitiesContainer" class="oj-flex-item">
            <h3 id="activitiesHeader">Activities</h3>
            <oj-list-view id="activitiesList" aria-
labelledby="activitiesHeader" data="[activityDataProvider]"
gridlines.item="visible">
                <template slot="itemTemplate">
                    <div>
                        <strong>
                            <oj-bind-text value="[[$current.data.name]]"></oj-bind-
text>
                        </strong>
                    </div>
                </template>
            </oj-list-view>
        </div>
        <!-- Parent Container 2 surrounds the Activity Items and Item
Details child containers -->
        <div id="parentContainer2" class="oj-flex oj-flex-item">
            <div id="activityItemsContainer" class="oj-flex-item"></div>
            <!-- Item Details container displays a chart upon selection -->
            <div id="itemDetailsContainer" class="oj-flex-item">
                <h3>Item Details</h3>
                <oj-label for="basicSelect">Select Chart:</oj-label>
                <oj-select-single id="basicSelect" class="selectSingleStyle"
data="[chartTypesDP]" value="{val}">
                    </oj-select-single>
                <oj-chart id="barChart" type="[val]"
data="[chartDataProvider]" animation-on-display="auto"
animation-on-data-change="auto" hover-behavior="dim"
class="chartStyle">
                    <template slot="itemTemplate" data-obj-as="item">
                        <oj-chart-item value="[item.data.value]" group-id="[
[item.data.group] ]"
series-id="[item.data.series]">
                            </oj-chart-item>

```

```

        </template>
      </oj-chart>
    </div>
  </div>
</div>
</div>

```

Task 4: Add Panels and Panel Color

1. Find the div element where id="parentContainer1", and then add two style classes to specify a panel with a border and background color.

```

<!-- The border and color for Parent Container 1 -->
<div
  id="parentContainer1"
  class="oj-flex oj-flex-init oj-panel oj-bg-warning-20"
></div>

```

2. Find the div element where id="activitiesContainer", and then add two style classes to specify a panel with a border and background color.

```

<!-- The border and color for the Activities container -->
<div id="activitiesContainer" class="oj-flex-item oj-bg-info-30">
  <h3 id="activitiesHeader">Activities</h3>
</div>

```

3. Find the div element where id="parentContainer2", and then add two style classes to specify a panel with a border and background color.

```

<!-- The border and color for Parent Container 2 -->
<div
  id="parentContainer2"
  class="oj-flex oj-flex-item oj-panel oj-bg-danger-30"
></div>

```

4. Find the div element where id="itemDetailsContainer", and then add two style classes to specify the panel and panel color.

```

<!-- The border and color for the Item Details container -->
<div
  id="itemDetailsContainer"
  class="oj-flex-item oj-panel oj-bg-neutral-30"
></div>

```

5. Save the dashboard.html file. Your code should look similar to the below code

```

<div class="oj-hybrid-padding">

```

```

<h1>Product Information</h1>
<!-- The border and color for Parent Container 1 -->
<div id="parentContainer1" class="oj-flex oj-flex-init oj-panel oj-bg-
warning-20">
  <!-- The border and color for the Activities container -->
  <div id="activitiesContainer" class="oj-flex-item oj-bg-info-30">
    <h3 id="activitiesHeader">Activities</h3>
    <oj-list-view id="activitiesList" aria-labelledby="activitiesHeader"
data="[[activityDataProvider]]"
    gridlines.item="visible">
      <template slot="itemTemplate">
        <div>
          <strong>
            <oj-bind-text value="[$current.data.name]"></oj-bind-text>
          </strong>
        </div>
      </template>
    </oj-list-view>
  </div>
  <!-- The border and color for Parent Container 2 -->
  <div id="parentContainer2" class="oj-flex oj-flex-item oj-panel oj-bg-
danger-30">
    <div id="activityItemsContainer" class="oj-flex-item"></div>
    <!-- The border and color for the Item Details container -->
    <div id="itemDetailsContainer" class="oj-flex-item oj-panel oj-bg-
neutral-30">
      <h3>Item Details</h3>
      <oj-label for="basicSelect">Select Chart:</oj-label>
      <oj-select-single id="basicSelect" class="selectSingleStyle"
data="[[chartTypesDP]]" value="{val}">
        </oj-select-single>
        <oj-chart id="barChart" type="[[val]]" data="[[chartDataProvider]]"
animation-on-display="auto"
          animation-on-data-change="auto" hover-behavior="dim"
class="chartStyle">
          <template slot="itemTemplate" data-obj-as="item">
            <oj-chart-item value="[[item.data.value]]" group-id="[[
[item.data.group] ]]"
              series-id="[[item.data.series]]">
            </oj-chart-item>
          </template>
        </oj-chart>
      </div>
    </div>
  </div>
</div>
</div>

```

Task 5: Run the Web App

npx ojet serve

Oracle JET tooling runs your web app in a local web browser where you can view the dashboard contents.

Product Information

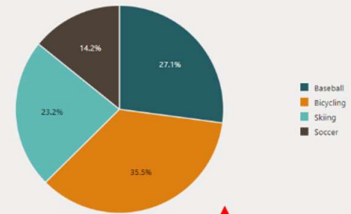
Activities

Baseball
Bicycling
Skiing
Soccer

Item Details

Select Chart:

Pie



Parent Container 1

Activities Container

Parent Container 2

Item Details Container